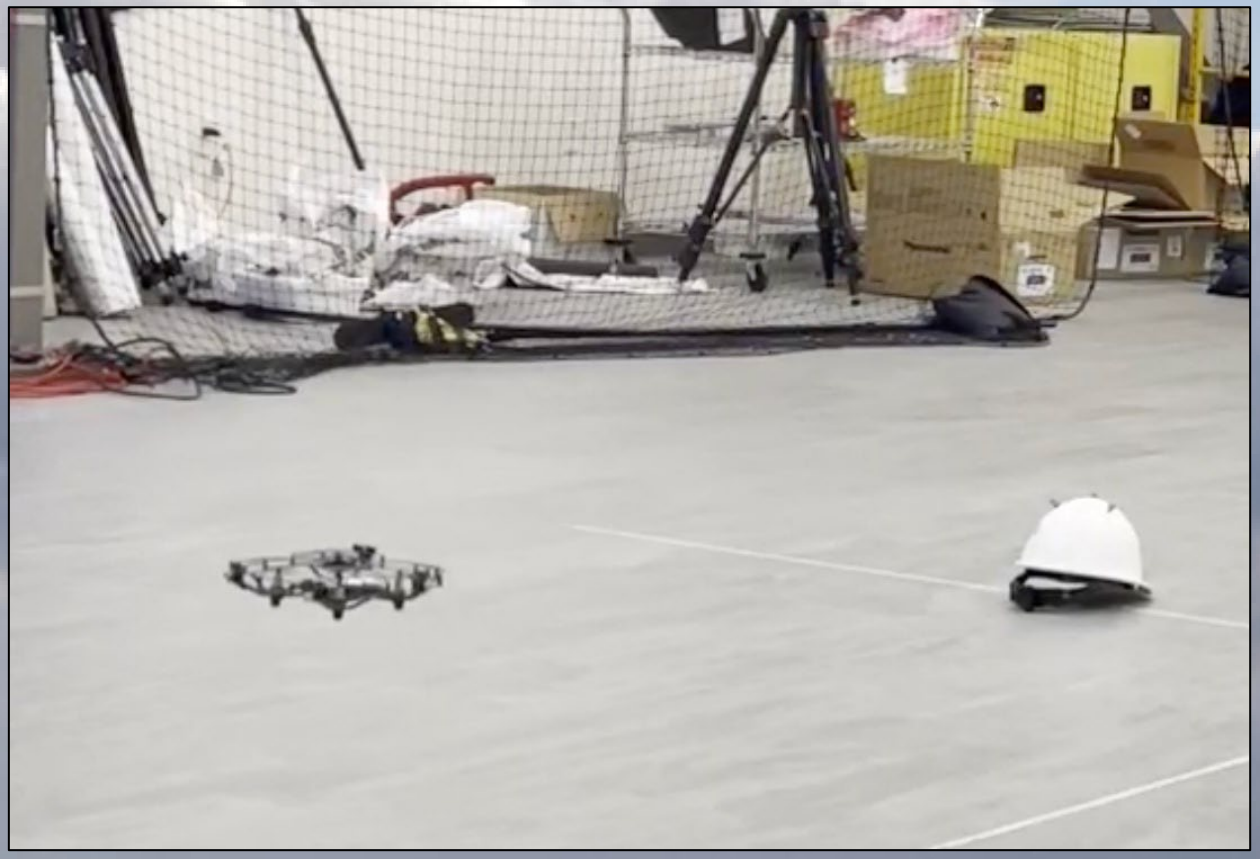
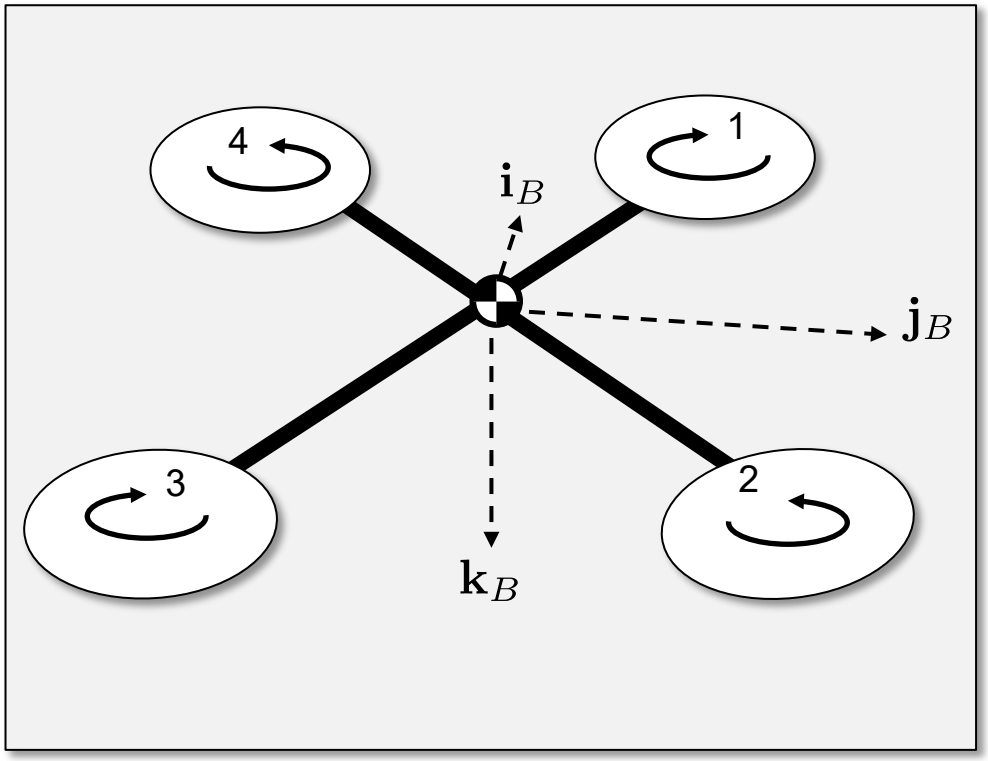
Quadrotor Dynamics and Control Review



ASEN 3801 Aerospace Vehicles Dynamics and Control Lab
UNIVERSITY OF COLORADO **BOULDER**

Multi-Rotor Dynamics

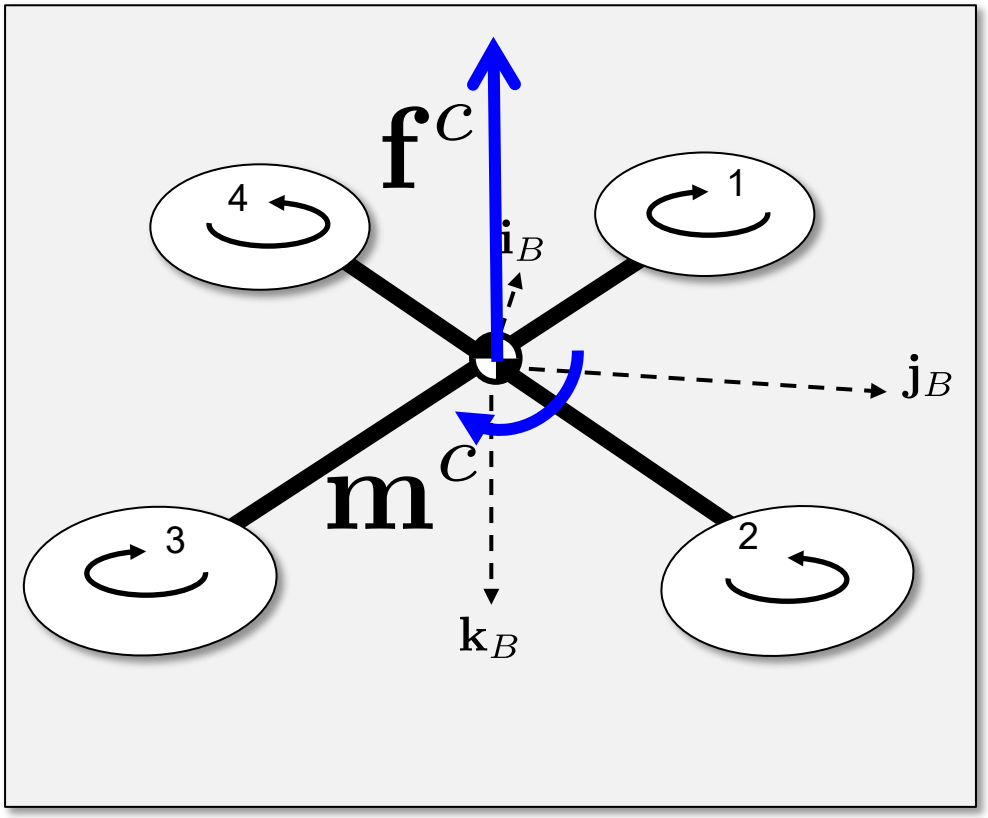
- Focus on quad-rotor design
  - Typical configuration
  - Balanced configuration



Control Force and Moment

- Thrust forces aligned with body z-axis

$$ {}^{c}\mathbf{f}_B = \begin{bmatrix} 0 \\ 0 \\ Z_c \end{bmatrix}, \; {}^{c}\mathbf{m}_B = \begin{bmatrix} L_c \\ M_c \\ N_c \end{bmatrix} $$

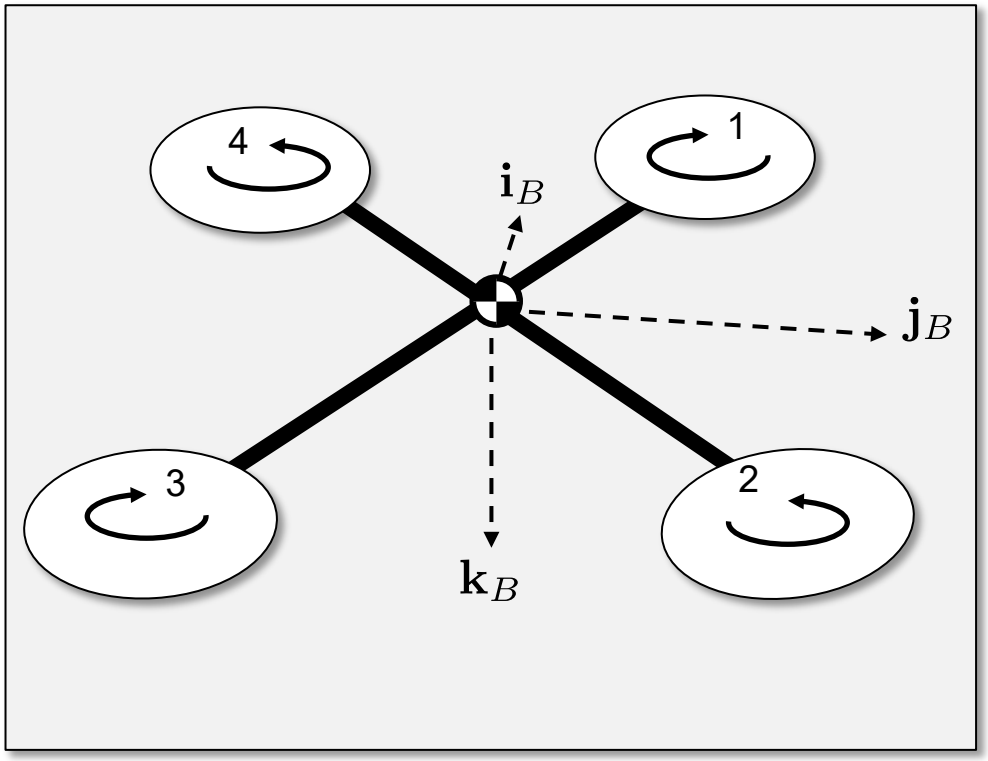


- Implications
  - No force in body x-axis or y-axis
  - Must tilt drone to move

Rotational Dynamics

Assume the aircraft is symmetric about the $\mathbf{i}_B - \mathbf{k}_B$ and $\mathbf{j}_B - \mathbf{k}_B$ planes, then $I_{xy} = I_{yz} = I_{xz} = 0$

$$\mathbf{I}_B = \begin{pmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{pmatrix}$$



Quadrotor Equations of Motion

$$\begin{pmatrix} \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \end{pmatrix} = \begin{pmatrix} \mathrm{c}_\theta \mathrm{c}_\psi & \mathrm{s}_\phi \mathrm{s}_\theta \mathrm{c}_\psi - \mathrm{c}_\phi \mathrm{s}_\psi & \mathrm{c}_\phi \mathrm{s}_\theta \mathrm{c}_\psi + \mathrm{s}_\phi \mathrm{s}_\psi \\ \mathrm{c}_\theta \mathrm{s}_\psi & \mathrm{s}_\phi \mathrm{s}_\theta \mathrm{s}_\psi + \mathrm{c}_\phi \mathrm{c}_\psi & \mathrm{c}_\phi \mathrm{s}_\theta \mathrm{s}_\psi - \mathrm{s}_\phi \mathrm{c}_\psi \\ -\mathrm{s}_\theta & \mathrm{s}_\phi \mathrm{c}_\theta & \mathrm{c}_\phi \mathrm{c}_\theta \end{pmatrix} \begin{pmatrix} u^E \\ v^E \\ w^E \end{pmatrix}$$

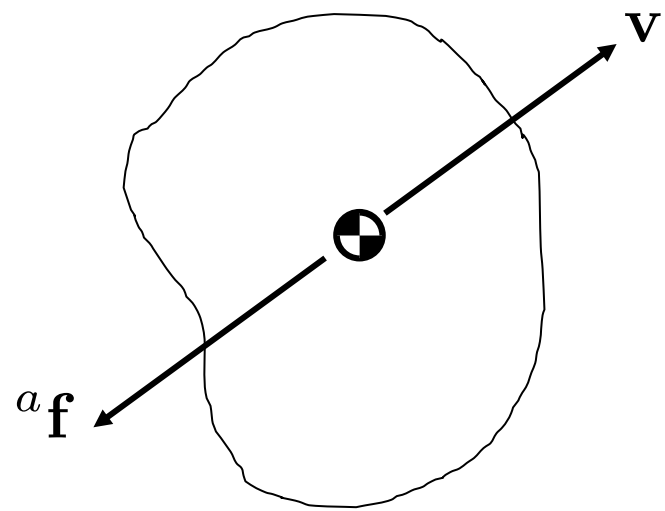
$$\begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix} = \begin{pmatrix} 1 & \sin\phi \tan\theta & \cos\phi \tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi \sec\theta & \cos\phi \sec\theta \end{pmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

$$\begin{pmatrix} \dot{u}^E \\ \dot{v}^E \\ \dot{w}^E \end{pmatrix} = \begin{pmatrix} rv^E - qw^E \\ pw^E - ru^E \\ qu^E - pv^E \end{pmatrix} + g \begin{pmatrix} -\sin\theta \\ \cos\theta \sin\phi \\ \cos\theta \cos\phi \end{pmatrix} + \frac{1}{m} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} + \frac{1}{m} \begin{pmatrix} 0 \\ 0 \\ Z_c \end{pmatrix}$$

$$\begin{pmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{pmatrix} = \begin{pmatrix} \frac{I_y - I_z}{I_x} qr \\ \frac{I_z - I_x}{I_y} pr \\ \frac{I_x - I_y}{I_z} pq \end{pmatrix} + \begin{pmatrix} \frac{1}{I_x} L \\ \frac{1}{I_y} M \\ \frac{1}{I_z} N \end{pmatrix} + \begin{pmatrix} \frac{1}{I_x} L_c \\ \frac{1}{I_y} M_c \\ \frac{1}{I_z} N_c \end{pmatrix}$$

Aerodynamic (Drag) Force

- Quadrotors typically fly with limited change in height so density constant
- Difficult to know drag coefficient and cross-section area



$$D = \underbrace{\frac{1}{2}\rho C_D A}_{\nu} V_a^2 = \nu V_a^2$$

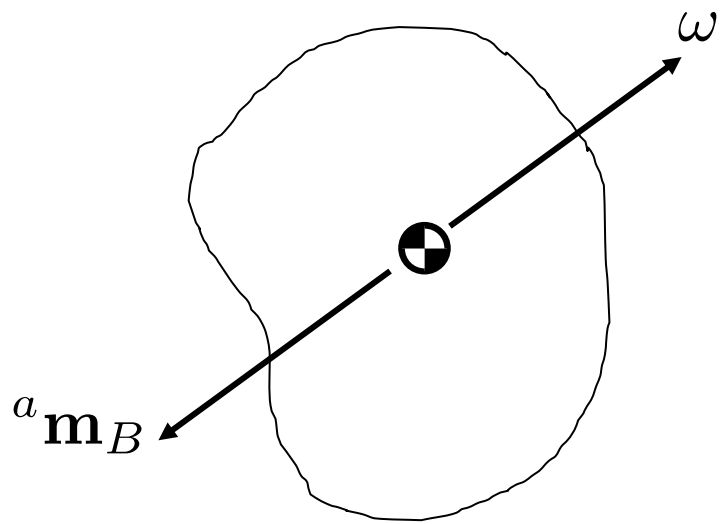
$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = {}^{a}\mathbf{f}_B = -D\frac{\mathbf{v}_B}{|\mathbf{v}|} = -\nu V_a^2 \frac{\mathbf{v_B}}{V_a}$$

$$= -\nu V_a \mathbf{v_B} = \boxed{-\nu V_a \begin{pmatrix} u \\ v \\ w \end{pmatrix}}$$

$$V_a = \sqrt{u^2 + v^2 + w^2}$$

Aerodynamic Moments

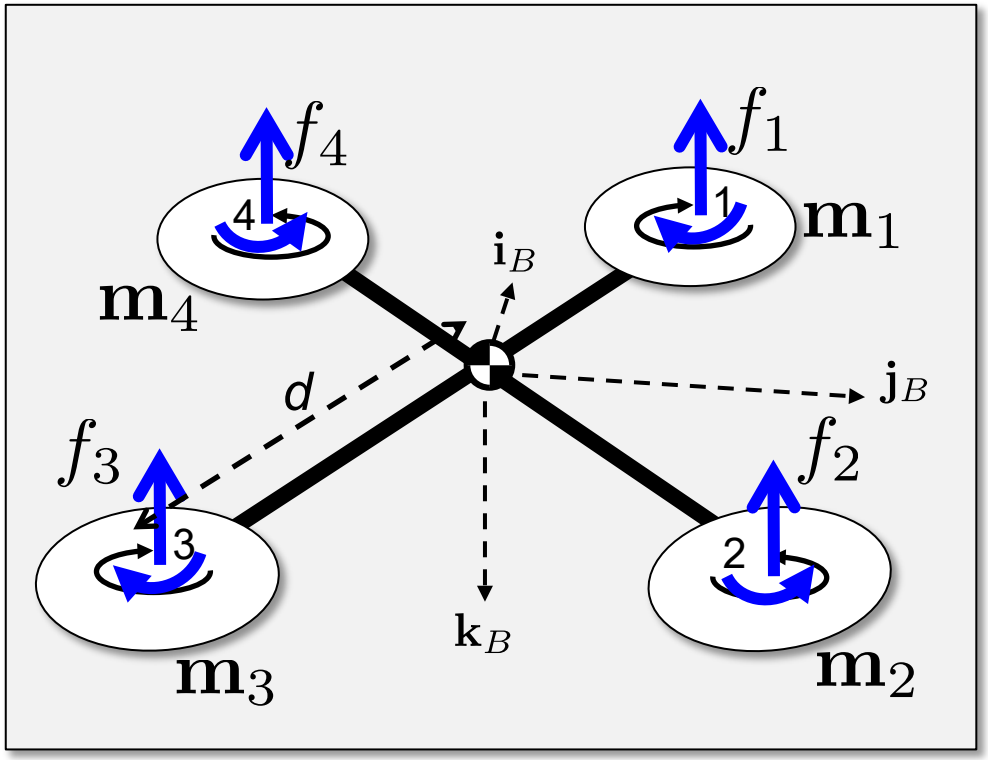
- Aerodynamic moment due to drag has same form as aerodynamic force



$$\begin{pmatrix} L \\ M \\ N \end{pmatrix} = {}^{a}\mathbf{m}_B = \boxed{-\mu\sqrt{p^2+q^2+r^2}\begin{pmatrix} p \\ q \\ r \end{pmatrix}}$$

Control Forces and Moments

- The control force and moments terms are linearly related to the motor forces, so can be written in matrix form:



$$\begin{bmatrix} Z_c \\ L_c \\ M_c \\ N_c \end{bmatrix} = \begin{bmatrix} -1 & -1 & -1 & -1 \\ -\frac{d}{\sqrt{2}} & -\frac{d}{\sqrt{2}} & \frac{d}{\sqrt{2}} & \frac{d}{\sqrt{2}} \\ \frac{d}{\sqrt{2}} & -\frac{d}{\sqrt{2}} & -\frac{d}{\sqrt{2}} & \frac{d}{\sqrt{2}} \\ k_m & -k_m & k_m & -k_m \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}$$

Quadrotor EOM Summary

$$\begin{pmatrix} \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \end{pmatrix} = \begin{pmatrix} \mathrm{c}_\theta \mathrm{c}_\psi & \mathrm{s}_\phi \mathrm{s}_\theta \mathrm{c}_\psi - \mathrm{c}_\phi \mathrm{s}_\psi & \mathrm{c}_\phi \mathrm{s}_\theta \mathrm{c}_\psi + \mathrm{s}_\phi \mathrm{s}_\psi \\ \mathrm{c}_\theta \mathrm{s}_\psi & \mathrm{s}_\phi \mathrm{s}_\theta \mathrm{s}_\psi + \mathrm{c}_\phi \mathrm{c}_\psi & \mathrm{c}_\phi \mathrm{s}_\theta \mathrm{s}_\psi - \mathrm{s}_\phi \mathrm{c}_\psi \\ -\mathrm{s}_\theta & \mathrm{s}_\phi \mathrm{c}_\theta & \mathrm{c}_\phi \mathrm{c}_\theta \end{pmatrix} \begin{pmatrix} u^E \\ v^E \\ w^E \end{pmatrix}$$

$$\begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix} = \begin{pmatrix} 1 & \sin\phi \tan\theta & \cos\phi \tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi \sec\theta & \cos\phi \sec\theta \end{pmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

$$\begin{pmatrix} \dot{u}^E \\ \dot{v}^E \\ \dot{w}^E \end{pmatrix} = \begin{pmatrix} rv^E - qw^E \\ pw^E - ru^E \\ qu^E - pv^E \end{pmatrix} + g \begin{pmatrix} -\sin\theta \\ \cos\theta \sin\phi \\ \cos\theta \cos\phi \end{pmatrix} + \frac{1}{m} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} + \frac{1}{m} \begin{pmatrix} 0 \\ 0 \\ Z_c \end{pmatrix}$$

$$\begin{pmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{pmatrix} = \begin{pmatrix} \frac{I_y - I_z}{I_x} qr \\ \frac{I_z - I_x}{I_y} pr \\ \frac{I_x - I_y}{I_z} pq \end{pmatrix} + \begin{pmatrix} \frac{1}{I_x} L \\ \frac{1}{I_y} M \\ \frac{1}{I_z} N \end{pmatrix} + \begin{pmatrix} \frac{1}{I_x} L_c \\ \frac{1}{I_y} M_c \\ \frac{1}{I_z} N_c \end{pmatrix}$$

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = -\nu V_a \begin{pmatrix} u \\ v \\ w \end{pmatrix}$$

$$\begin{bmatrix} Z_c \\ L_c \\ M_c \\ N_c \end{bmatrix} = \begin{bmatrix} -1 & -1 & -1 & -1 \\ -\frac{d}{\sqrt{2}} & -\frac{d}{\sqrt{2}} & \frac{d}{\sqrt{2}} & \frac{d}{\sqrt{2}} \\ \frac{d}{\sqrt{2}} & -\frac{d}{\sqrt{2}} & -\frac{d}{\sqrt{2}} & \frac{d}{\sqrt{2}} \\ k_m & -k_m & k_m & -k_m \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}$$

$$\begin{pmatrix} L \\ M \\ N \end{pmatrix} = -\mu \sqrt{p^2 + q^2 + r^2} \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

Trim Condition

In a "**steady hover**" equilibrium condition

$$\dot{x}_{E,0} = 0 \qquad u_0^E = 0 \qquad \dot{u}_0^E = 0$$
$$\dot{y}_{E,0} = 0 \qquad v_0^E = 0 \qquad \dot{v}_0^E = 0$$
$$\dot{z}_{E,0} = 0 \qquad w_0^E = 0 \qquad \dot{w}_0^E = 0$$

Subscript 0 denotes trim value

$$p_0 = 0 \qquad \dot{p}_0 = 0 \qquad \phi_0 = 0$$
$$q_0 = 0 \qquad \dot{q}_0 = 0 \qquad \theta_0 = 0$$
$$r_0 = 0 \qquad \dot{r}_0 = 0 \qquad \psi_0 = \cdot$$

Any constant value

Linearized Quadrotor EOM

$$\begin{pmatrix} \Delta\dot{x}_E \\ \Delta\dot{y}_E \\ \Delta\dot{z}_E \end{pmatrix} = \begin{pmatrix} \Delta u \\ \Delta v \\ \Delta w \end{pmatrix}$$

$$\begin{pmatrix} \Delta\dot{\phi} \\ \Delta\dot{\theta} \\ \Delta\dot{\psi} \end{pmatrix} = \begin{pmatrix} \Delta p \\ \Delta q \\ \Delta r \end{pmatrix}$$

$$\begin{pmatrix} \Delta\dot{u} \\ \Delta\dot{v} \\ \Delta\dot{w} \end{pmatrix} = g \begin{pmatrix} -\Delta\theta \\ \Delta\phi \\ 0 \end{pmatrix} + \frac{1}{m} \begin{pmatrix} 0 \\ 0 \\ \Delta Z_c \end{pmatrix}$$

$$\begin{pmatrix} \Delta\dot{p} \\ \Delta\dot{q} \\ \Delta\dot{r} \end{pmatrix} = \begin{pmatrix} \frac{1}{I_x}\Delta L_c \\ \frac{1}{I_y}\Delta M_c \\ \frac{1}{I_z}\Delta N_c \end{pmatrix}$$

Linearized Quadrotor EOM

Longitudinal

$$\begin{pmatrix} \Delta\dot{x}_E \\ \Delta\dot{u} \\ \Delta\dot{\theta} \\ \Delta\dot{q} \end{pmatrix} = \begin{pmatrix} \Delta u \\ -g\Delta\theta \\ \Delta q \\ \frac{1}{I_y}\Delta M_c \end{pmatrix}$$

Lateral

$$\begin{pmatrix} \Delta\dot{y}_E \\ \Delta\dot{v} \\ \Delta\dot{\phi} \\ \Delta\dot{p} \end{pmatrix} = \begin{pmatrix} \Delta v \\ g\Delta\phi \\ \Delta p \\ \frac{1}{I_x}\Delta L_c \end{pmatrix}$$

Vertical

$$\begin{pmatrix} \Delta\dot{z}_E \\ \Delta\dot{w} \end{pmatrix} = \begin{pmatrix} \Delta w \\ \frac{1}{m}\Delta Z_c \end{pmatrix}$$

Spin

$$\begin{pmatrix} \Delta\dot{\psi} \\ \Delta\dot{r} \end{pmatrix} = \begin{pmatrix} \Delta r \\ \frac{1}{I_z}\Delta N_c \end{pmatrix}$$

Feedback control

Let's consider a **feedback control law**

$$\Delta L_c = -k_1 \Delta p - k_2 \Delta \phi$$

which we can implement by measuring the roll rate and roll angle and "feeding them back" into the command for the roll moment.

Closed Loop Behavior

- Consider the feedback law

$$\Delta L_c = -k_1 \Delta p - k_2 \Delta \phi$$

- There are two ways mathematically to describe the resulting "closed loop behavior"

$$\begin{pmatrix} \Delta\dot{\phi} \\ \Delta\dot{p} \end{pmatrix} = \begin{pmatrix} \Delta p \\ \frac{1}{I_x}\Delta L_c \end{pmatrix}$$

$$\Delta\ddot{\phi} = \Delta\dot{p} = \frac{1}{I_x}\Delta L_c$$

$$\Delta\ddot{\phi} = \frac{1}{I_x}\left(-k_1\Delta p - k_2\Delta\phi\right)$$

$$\Delta\ddot{\phi} + \frac{k_1}{I_x}\Delta\dot{\phi} + \frac{k_2}{I_x}\Delta\phi = 0$$

$$\begin{pmatrix} \Delta\dot{\phi} \\ \Delta\dot{p} \end{pmatrix} = \begin{pmatrix} \Delta p \\ \frac{1}{I_x}\Delta L_c \end{pmatrix}$$

$$\begin{pmatrix} \Delta\dot{\phi} \\ \Delta\dot{p} \end{pmatrix} = \begin{pmatrix} \Delta p \\ \frac{1}{I_x}\left(-k_1\Delta p - k_2\Delta\phi\right) \end{pmatrix}$$

$$\begin{pmatrix} \Delta\dot{\phi} \\ \Delta\dot{p} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \frac{-k_2}{I_x} & \frac{-k_1}{I_x} \end{pmatrix} \begin{pmatrix} \Delta\phi \\ \Delta p \end{pmatrix}$$

Mathematically equivalent!

Closed Loop Control Example: Lateral Dynamics

This is an example. It IS NOT the equation to implement for this lab. You must determine that for the assignment.

$$\begin{pmatrix} \Delta\dot{y}_E \\ \Delta\dot{v} \\ \Delta\dot{\phi} \\ \Delta\dot{p} \end{pmatrix} = \begin{pmatrix} \Delta v \\ g\Delta\phi \\ \Delta p \\ \frac{1}{I_x}\Delta L_c \end{pmatrix}$$

Linear aircraft model = "the plant"

"Control law"

+

$$\Delta L_c = -k_1\Delta p - k_2\Delta\phi - k_3\Delta v - k_3k_4\Delta y_E + k_3k_4\Delta y_{E,r}$$

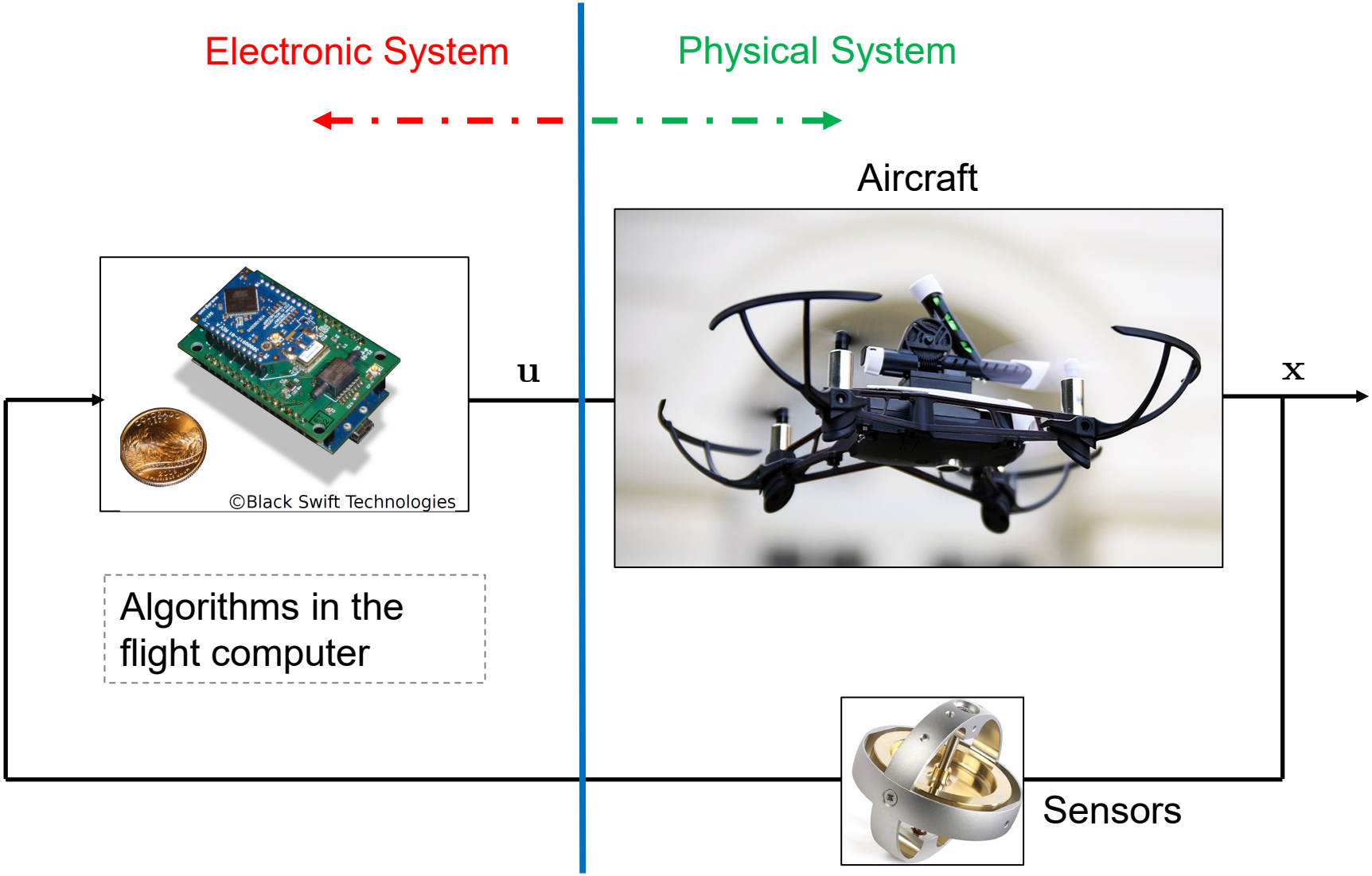
=

"Closed loop system"

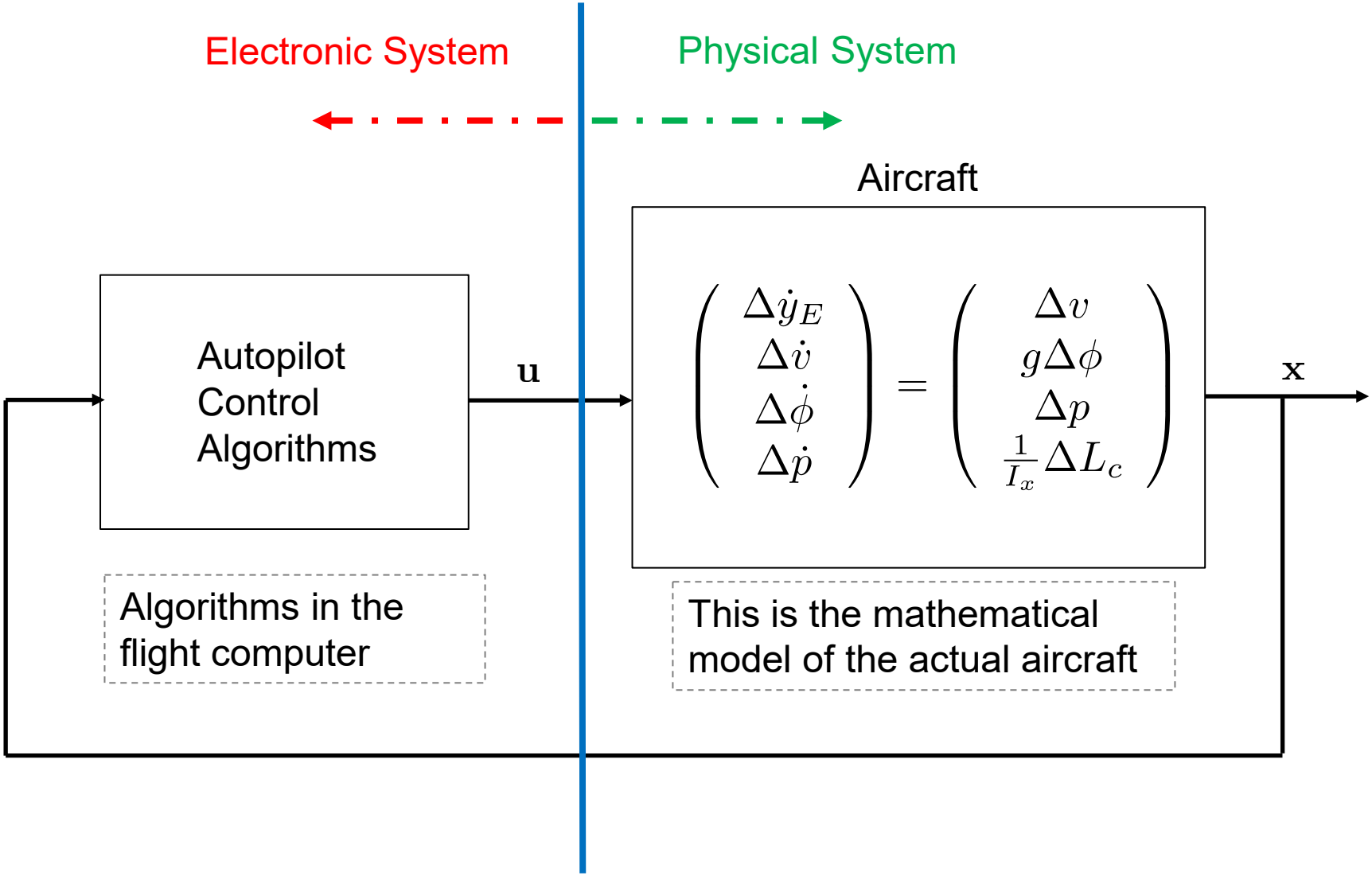
$$\begin{pmatrix} \Delta\dot{y}_E \\ \Delta\dot{v} \\ \Delta\dot{\phi} \\ \Delta\dot{p} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & g & 0 \\ 0 & 0 & 0 & 1 \\ \frac{-k_3k_4}{I_x} & \frac{-k_3}{I_x} & \frac{-k_2}{I_x} & \frac{-k_1}{I_x} \end{pmatrix} \begin{pmatrix} \Delta y_E \\ \Delta v \\ \Delta\phi \\ \Delta p \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{k_3k_4}{I_x} \end{pmatrix} \Delta y_{E,r}$$

Design EXAMPLE

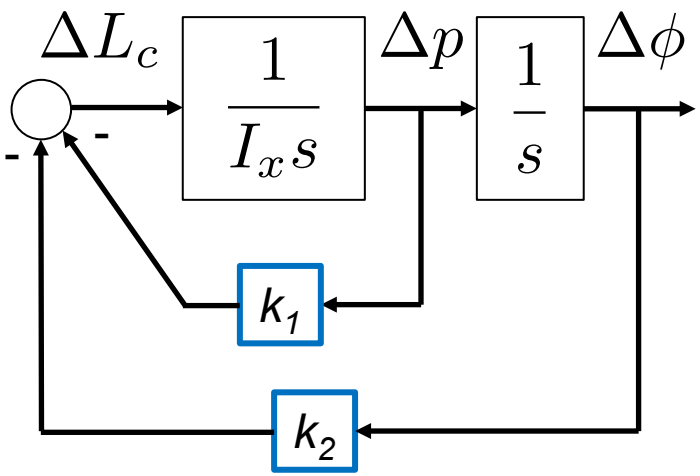
Feedback Control



Feedback Control



Inner Loop Behavior



$$\Delta\ddot{\phi} + \frac{k_1}{I_x}\Delta\dot{\phi} + \frac{k_2}{I_x}\Delta\phi = 0$$

$$\Downarrow$$

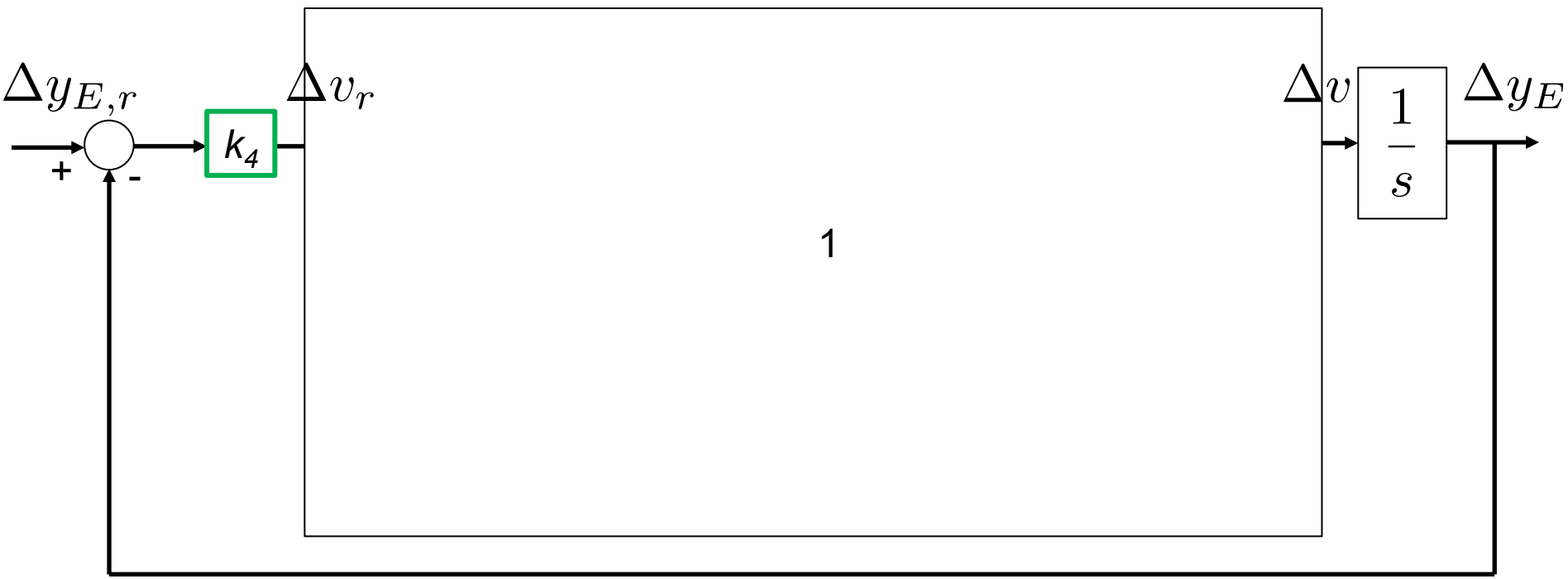
$$\Delta\ddot{\phi} + 2\zeta\omega_n\Delta\dot{\phi} + \omega_n^2\Delta\phi = 0$$

$$\boxed{\omega_n = \sqrt{\frac{k_2}{I_x}} \qquad \zeta = \frac{k_1}{2\sqrt{k_2 I_x}}}$$

which has roots $\lambda = -\zeta\omega_n \pm i\omega_n\sqrt{1-\zeta^2}$

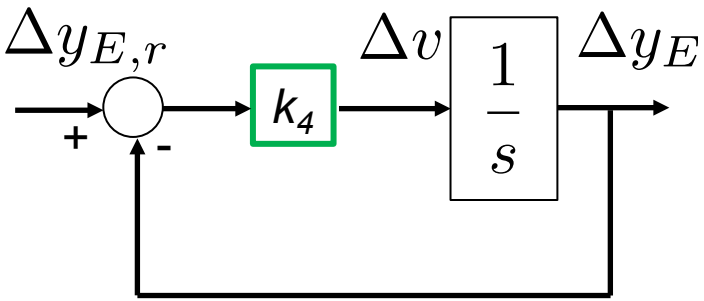
1. Design *inner loop* (k_1 and k_2) to get desired, “fast”, response. Use either Laplace domain or state space approach.

Design Strategy



2. Design *outer loop* (k_4) as if you get a perfect response from the inner loop
   - time constant should be an order of magnitude slower than period of inner loop

Design Strategy



$$\Longleftrightarrow \quad \Delta \dot{y}_E = k_4 \left(\Delta y_{E,r} - \Delta y_E \right)$$

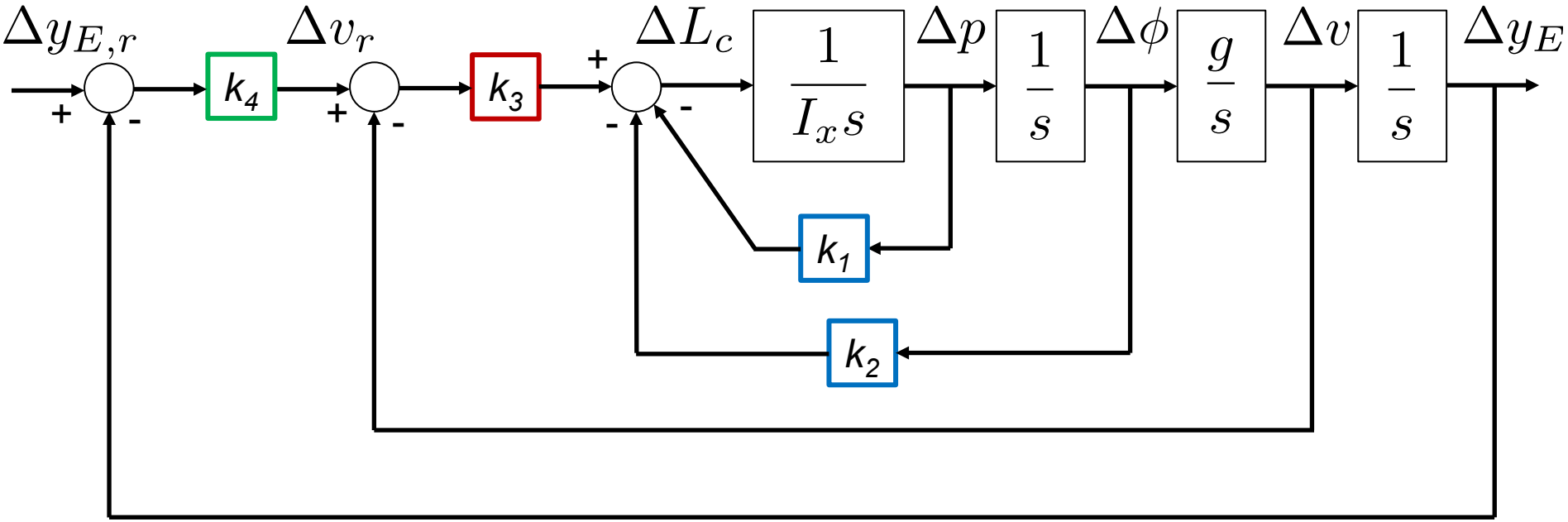
$$\Downarrow$$

$$\Delta y_E(t) = \Delta y_{E,r} \left(1 - e^{-k_4 t} \right)$$

$$\Downarrow$$

Time constant $\tau = \dfrac{1}{k_4} > 10 \cdot \dfrac{2\pi}{\zeta \omega_n}$

Design Strategy



3. Design middle loop (k_3) to get overall performance

Closed Loop State Space Model

$$\begin{pmatrix} \Delta\dot{y}_E \\ \Delta\dot{v} \\ \Delta\dot{\phi} \\ \Delta\dot{p} \end{pmatrix} = \begin{pmatrix} \Delta v \\ g\Delta\phi \\ \Delta p \\ \frac{1}{I_x}\Delta L_c \end{pmatrix}$$

+

$$\Delta L_c = -k_1\Delta p - k_2\Delta\phi - k_3\Delta v - k_3k_4\Delta y_E + k_3k_4\Delta y_{E,r}$$

=

$$\begin{pmatrix} \Delta\dot{y}_E \\ \Delta\dot{v} \\ \Delta\dot{\phi} \\ \Delta\dot{p} \end{pmatrix} = \boxed{\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & g & 0 \\ 0 & 0 & 0 & 1 \\ \frac{-k_3k_4}{I_x} & \frac{-k_3}{I_x} & \frac{-k_2}{I_x} & \frac{-k_1}{I_x} \end{pmatrix}} \begin{pmatrix} \Delta y_E \\ \Delta v \\ \Delta\phi \\ \Delta p \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{k_3k_4}{I_x} \end{pmatrix} \Delta y_{E,r}$$

Look at the eigenvalues of this matrix.

Design Strategy

$$\begin{pmatrix} \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \end{pmatrix} = \begin{pmatrix} \mathrm{c}_\theta \mathrm{c}_\psi & \mathrm{s}_\phi \mathrm{s}_\theta \mathrm{c}_\psi - \mathrm{c}_\phi \mathrm{s}_\psi & \mathrm{c}_\phi \mathrm{s}_\theta \mathrm{c}_\psi + \mathrm{s}_\phi \mathrm{s}_\psi \\ \mathrm{c}_\theta \mathrm{s}_\psi & \mathrm{s}_\phi \mathrm{s}_\theta \mathrm{s}_\psi + \mathrm{c}_\phi \mathrm{c}_\psi & \mathrm{c}_\phi \mathrm{s}_\theta \mathrm{s}_\psi - \mathrm{s}_\phi \mathrm{c}_\psi \\ -\mathrm{s}_\theta & \mathrm{s}_\phi \mathrm{c}_\theta & \mathrm{c}_\phi \mathrm{c}_\theta \end{pmatrix} \begin{pmatrix} u^E \\ v^E \\ w^E \end{pmatrix}$$

$$\begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix} = \begin{pmatrix} 1 & \sin\phi \tan\theta & \cos\phi \tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi \sec\theta & \cos\phi \sec\theta \end{pmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

$$\begin{pmatrix} \dot{u}^E \\ \dot{v}^E \\ \dot{w}^E \end{pmatrix} = \begin{pmatrix} rv^E - qw^E \\ pw^E - ru^E \\ qu^E - pv^E \end{pmatrix} + g \begin{pmatrix} -\sin\theta \\ \cos\theta \sin\phi \\ \cos\theta \cos\phi \end{pmatrix} + \frac{1}{m} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} + \frac{1}{m} \begin{pmatrix} 0 \\ 0 \\ Z_c \end{pmatrix}$$

$$\begin{pmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{pmatrix} = \begin{pmatrix} \frac{I_y - I_z}{I_x} qr \\ \frac{I_z - I_x}{I_y} pr \\ \frac{I_x - I_y}{I_z} pq \end{pmatrix} + \begin{pmatrix} \frac{1}{I_x} L \\ \frac{1}{I_y} M \\ \frac{1}{I_z} N \end{pmatrix} + \begin{pmatrix} \frac{1}{I_x} \boxed{L_c} \\ \frac{1}{I_y} M_c \\ \frac{1}{I_z} N_c \end{pmatrix}$$

4. Verify with full nonlinear equations

$$\Delta L_c = -k_1 \Delta p - k_2 \Delta \phi - k_3 \Delta v - k_3 k_4 \Delta y_E + k_3 k_4 \Delta y_{E,r}$$

Overall Design Strategy

Longitudinal

$$\begin{pmatrix} \Delta\dot{x}_E \\ \Delta\dot{u} \\ \Delta\dot{\theta} \\ \Delta\dot{q} \end{pmatrix} = \begin{pmatrix} \Delta u \\ -g\Delta\theta \\ \Delta q \\ \frac{1}{I_y}\Delta M_c \end{pmatrix}$$

Lateral

$$\begin{pmatrix} \Delta\dot{y}_E \\ \Delta\dot{v} \\ \Delta\dot{\phi} \\ \Delta\dot{p} \end{pmatrix} = \begin{pmatrix} \Delta v \\ g\Delta\phi \\ \Delta p \\ \frac{1}{I_x}\Delta L_c \end{pmatrix}$$

Vertical

$$\begin{pmatrix} \Delta\dot{z}_E \\ \Delta\dot{w} \end{pmatrix} = \begin{pmatrix} \Delta w \\ \frac{1}{m}\Delta Z_c \end{pmatrix}$$

Spin

$$\begin{pmatrix} \Delta\dot{\psi} \\ \Delta\dot{r} \end{pmatrix} = \begin{pmatrix} \Delta r \\ \frac{1}{I_z}\Delta N_c \end{pmatrix}$$

Overall Design Strategy

Same process for longitudinal

Longitudinal

$$\begin{pmatrix} \Delta\dot{x}_E \\ \Delta\dot{u} \\ \Delta\dot{\theta} \\ \Delta\dot{q} \end{pmatrix} = \begin{pmatrix} \Delta u \\ -g\Delta\theta \\ \Delta q \\ \frac{1}{I_y}\Delta M_c \end{pmatrix}$$

Lateral

$$\begin{pmatrix} \Delta\dot{y}_E \\ \Delta\dot{v} \\ \Delta\dot{\phi} \\ \Delta\dot{p} \end{pmatrix} = \begin{pmatrix} \Delta v \\ g\Delta\phi \\ \Delta p \\ \frac{1}{I_x}\Delta L_c \end{pmatrix}$$

Vertical

$$\begin{pmatrix} \Delta\dot{z}_E \\ \Delta\dot{w} \end{pmatrix} = \begin{pmatrix} \Delta w \\ \frac{1}{m}\Delta Z_c \end{pmatrix}$$

Spin

$$\begin{pmatrix} \Delta\dot{\psi} \\ \Delta\dot{r} \end{pmatrix} = \begin{pmatrix} \Delta r \\ \frac{1}{I_z}\Delta N_c \end{pmatrix}$$

Overall Design Strategy

Same process for longitudinal

Longitudinal

$$\begin{pmatrix} \Delta\dot{x}_E \\ \Delta\dot{u} \\ \Delta\dot{\theta} \\ \Delta\dot{q} \end{pmatrix} = \begin{pmatrix} \Delta u \\ -g\Delta\theta \\ \Delta q \\ \frac{1}{I_y}\Delta M_c \end{pmatrix}$$

Lateral

$$\begin{pmatrix} \Delta\dot{y}_E \\ \Delta\dot{v} \\ \Delta\dot{\phi} \\ \Delta\dot{p} \end{pmatrix} = \begin{pmatrix} \Delta v \\ g\Delta\phi \\ \Delta p \\ \frac{1}{I_x}\Delta L_c \end{pmatrix}$$

Vertical

$$\begin{pmatrix} \Delta\dot{z}_E \\ \Delta\dot{w} \end{pmatrix} = \begin{pmatrix} \Delta w \\ \frac{1}{m}\Delta Z_c \end{pmatrix}$$

Spin

$$\begin{pmatrix} \Delta\dot{\psi} \\ \Delta\dot{r} \end{pmatrix} = \begin{pmatrix} \Delta r \\ \frac{1}{I_z}\Delta N_c \end{pmatrix}$$

Vertical and spin both second order systems,
so design like the inner loops